Example 4.6

Find (i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ (ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)$ (iii) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)$

Solution

It is known that $\cos^{-1}x:[-1,1]\to[0,\pi]$ is given by

$\cos^{-1}x=y$ if and only if $x=\cos y$ for $-1\le x\le 1$ and $0\le y\le\pi$.

Thus, we have

(i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)=\frac{3\pi}{4}$, since $\frac{3\pi}{4}\in[0,\pi]$ and $\cos\frac{3\pi}{4}=\cos\left(\pi-\frac{\pi}{4}\right)=-\cos\frac{\pi}{4}=-\frac{1}{\sqrt{2}}$.

(ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)=\cos^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right)=\frac{\pi}{3}$, since $-\frac{\pi}{3}\notin[0,\pi]$, but $\frac{\pi}{3}\in[0,\pi]$.

(iii) $\cos^{-1}\left(\cos\left(\frac{7\pi}{6}\right)\right)=\frac{5\pi}{6}$, since $\cos\left(\frac{7\pi}{6}\right)=\cos\left(\pi+\frac{\pi}{6}\right)=-\frac{\sqrt{3}}{2}=\cos\left(\frac{5\pi}{6}\right)$ and $\frac{5\pi}{6}\in[0,\pi]$.

Example 4.7

Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$.

Solution

By definition, the domain of $y=\cos^{-1}x$ is $-1\le x\le 1$ or $|x|\le 1$. This leads to

$-1\le\frac{2+\sin x}{3}\le 1$ which is same as $-3\le 2+\sin x\le 3$.

So, $-5\le\sin x\le 1$ reduces to $-1\le\sin x\le 1$, which gives

$-\sin^{-1}(1)\le x\le\sin^{-1}(1)$ or $-\frac{\pi}{2}\le x\le\frac{\pi}{2}$.

Thus, the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$ is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

EXERCISE 4.2

1. Find all values of x such that

   (i) $-6\pi\le x\le 6\pi$ and $\cos x=0$ (ii) $-5\pi\le x\le 5\pi$ and $\cos x=1$.

2. State the reason for $\cos^{-1}\left[\cos\left(-\frac{\pi}{6}\right)\right]\ne-\frac{\pi}{6}$.

3. Is $\cos^{-1}(-x)=\pi-\cos^{-1}(x)$ true? Justify your answer.

4. Find the principal value of $\cos^{-1}\left(\frac{1}{2}\right)$.

5. Find the value of

   (i) $2\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}\left(\frac{1}{2}\right)$ (ii) $\cos^{-1}\left(\frac{1}{2}\right)+\sin^{-1}(-1)$

   (iii) $\cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17}-\sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$.

Chapter 4 Inverse Trigonometry.indd 142 31-01-2020 17:43:30